

PrepPilot — Agent System & Handler Library

8 specialised agents · 417 ready-made handlers · isolated Python sandbox

Multi-Agent Orchestration

8 agents coordinated by DS-Agent · 4 core pipeline (top row) + 4 slash-command specialists (bottom row)



DS-Agent
Orchestrator · main /chat endpoint
routes incoming message



Planner
Two-stage router · builds JSON plan
narrows 417 handlers → ~150



Step Executor
Handler resolver · codegen for
auto-retry once on error



Train Agent
Auto ML training pipeline
/train · 27 algos · Optuna tuning



Feature Agent
Automated feature engineering
/feature · encoders · scalers · PCA



Auto-Clean
AI plans + applies cleaning steps
/cleaning · report card

use

Handler Registry · 417 ready-made operations

One file per handler · registered through HANDLER_REGISTRY · grouped by 7 categories

stats
56
describe · correlation · chi²

clean
60
fill_nulls · SMOTE · dedupe

transform
64
pivot · encode · sample

viz
56
bar · scatter · heatmap

feature
63
PCA · RFE · rollouts

run in

Sandboxed exec() · isolated Python runtime

Every handler & codegen step runs here · safe namespace · pre-loaded libraries · Plotly theme

EXECUTION CONTROLS

 **Restricted namespace**

 **Auto-retry once on error**

60s **Sandbox timeout**

Aa **Theme**

PRE-INSTALLED LIBRARIES

pandas

numpy

scikit-learn

xgboost

lightgbm

optuna

 Pipeline tier  Uses / runs in